

Study Questions: Systems

1. What distinguishes a "system" from a mere collection of parts, and how does Active Inference formalize this distinction?
2. Explain the concept of a Markov Blanket in non-mathematical language. What philosophical work does it do?
3. How does Merleau-Ponty's phenomenology of the body challenge naive conceptions of system boundaries?
4. Define autopoiesis as introduced by Maturana and Varela. Why is circular causality central to their account?
5. In what sense does the Free Energy Principle extend the autopoietic account of living systems?
6. What does it mean to say that "to exist is to resist dissolution"? Is this a metaphysical claim, a tautology, or an empirical hypothesis?
7. State the life-mind continuity thesis. What evidence or arguments support it?
8. What is the "fallacy of equivocation" objection to the life-mind continuity thesis, and how might a proponent respond?
9. How does the Markov Blanket formalism relate to the philosophical concept of individuality?
10. Can non-living systems (hurricanes, candle flames) have Markov Blankets? What are the philosophical consequences?
11. How does the Ship of Theseus paradox look different from a Markov Blanket perspective compared to a substance-based ontology?
12. Evaluate Clark and Chalmers' Extended Mind Hypothesis in light of the Markov Blanket framework. Does the framework support or undermine it?
13. What is the difference between an "epistemic" boundary (statistical separation) and an "ontic" boundary (physical separation)? Which does Active Inference invoke?
14. How does the concept of sensory and active states in a Markov Blanket relate to the philosophical distinction between receptivity and spontaneity?
15. Friston argues that any system that persists must minimize variational free energy. Is this a descriptive claim about nature or a definitional condition?

16. What role does the concept of "surprise" play in the systems-level account? How does it differ from the everyday meaning of the word?
17. Compare the Active Inference account of system boundaries with Gibson's ecological psychology. Are they compatible?
18. How might a process philosopher (e.g., Whitehead) view the Markov Blanket formalism? Is it consistent with a process-relational ontology?
19. If systems are defined by their boundaries, what happens at the boundaries of boundaries? Is there an infinite regress problem?
20. Construct a philosophical argument for or against the claim that the Free Energy Principle is a genuine *law of nature* rather than a useful modeling convention.